

SK Matrix-Model Contributions to the Squared Commutator

External assignment (by external-propagator appearance): $\phi_r^1(x_1) \phi_a^1(x_2) \phi_r^2(x_3) \phi_a^2(x_4)$, with $x_1 = (t, \mathbf{x})$, $x_2 = (0, \mathbf{0})$, $x_3 = (t, \mathbf{x})$, $x_4 = (0, \mathbf{0})$. Propagator notation: $G_{\ell_1 \ell_2}^{c_1 c_2}(z, y)$ (upper indices: time folds, lower indices: r/a labels). Translation table used: same-fold $G_{rr} \rightarrow G_K = \frac{1}{2}(G^> + G^<)$, $G_{ra} \rightarrow G_R$, $G_{ar} \rightarrow G_A$, $G_{aa} \rightarrow 0$; cross-fold $G_{rr}^{12} \rightarrow G^<$, $G_{rr}^{21} \rightarrow G^>$, and any cross-fold propagator with an a endpoint vanishes. Contour ordering convention: contour (2) is later than contour (1), matching `contour_ra_translation.txt`. Each term carries explicit net matrix-size factor N_c^{p-4} (raw loop count p minus the overall $1/N^4$ normalization).

Topology 2

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.

Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.

Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_1 = 128ig_2^3 N_c^{-4} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(x_1, z_1) G_R(x_3, z_3) G_R(z_2, z_1)$$

$$\mathcal{T}_2 = 128ig_2^3 N_c^{-4} G^<(z_1, z_3) G^<(z_2, z_3) G_A(x_2, z_2) G_A(x_4, z_3) G_A(z_2, z_1) G_K(z_2, z_1) G_R(x_1, z_1) G_R(x_3, z_3)$$